STAT2004 – Tutorial Week 11

Exercises 2–3 covered in the tutorial. Please try the other exercises at home. You have tackled similar problems to Exercise 4 in first-year courses such as STAT1201/1301, but without a rigorous justification.

Exercise 1 (Basic properties of the multinomial distribution)

In the lectures we introduced the *multinomial distribution* as a generalization of the binomial distribution when there more than two categories. This question looks at some aspects of this generalization.

Let $\boldsymbol{X} = (X_1, X_2, \ldots, X_K)^\top$ be a random sample of counts from a multinomial distribution with parameters $\boldsymbol{p} = (p_1, p_2, \ldots, p_K)^\top$, such that $1 = p_1 + p_2 + \ldots + p_K$ and $n = X_1 + X_2 + \ldots + X_K$ is the sample size. Recall that the joint pmf of $\boldsymbol{X}$ is given by

$$P\big(X_1 = x_1, X_2 = x_2, \ldots, X_K = x_k\big) = \frac{n!}{x_1! x_2! \cdots x_K!}\, p_1^{x_1}\, p_2^{x_2} \cdots p_K^{x_K} \ ,$$

for vectors $\boldsymbol{x} = (x_1, x_2, \ldots, x_K)^\top$ that sum to n .

(a) Show that the *marginal* distribution of each count X_k is Binomial(n, p_k).

(b) Hence, show that the expected value and variance of each count X_k are given by $E(X_k) = np_k$ and $\mathrm{Var}(X_k) = np_k(1 - p_k)$, respectively.

(c) Clearly, the counts X_k and $X_{k'}$ in any two categories k and k' are dependent (why?). Show that the covariance between two counts X_k and $X_{k'}$ is given by

$$\mathrm{Cov}\,(X_k, X_{k'}) = -np_k\, p_{k'} \ , \quad \text{for } k \neq k' \ .$$

(d) For $K = 2$ categories, what is the correlation between the counts X_1 and X_2 in the two categories? (Do we really need a formula to deduce this?)

Exercise 2 (Goodness-of-fit test for a die)

In Tutorial 9 Exercise 4, recall that a die was rolled 100 times and the number of "sixes" was observed to be 25. The full set of observed counts from this experiment is given below:

	1	2	3	4	5	6
counts	17	15	12	18	13	25

We are interested in testing whether this die is fair or loaded.

(a) Write down the null and alternative hypotheses, both *in words* and *mathematically* in terms of the underlying probability vector $\boldsymbol{p} = (p_1, p_2, p_3, p_4, p_5, p_6)^\top$.

(b) If the null hypothesis is indeed true, what counts would we have expected to see for each side of this die?
(c) Compute the (i) generalized likelihood ratio Λ, (ii) the transformation $-2\log\Lambda$, and (iii) Pearson's χ^2 test statistics for these data.
(d) What is the distribution of Pearson's χ^2 statistic under the null hypothesis?
(e) Compute and interpret a p-value for testing the fairness of this die. What do you conclude?

In Tutorial 8 Exercise 3, recall that we found *moderately strong* evidence that the proportion of "sixes" p_6 is not equal to 1/6 for this die (with an associated p-value of 0.043).

(f) How do you reconcile the apparent contradiction with your conclusions from part (e)?
(g) Suggest a quick way to correct the p-value of 0.043 from Tutorial 8 Exercise 3. Do your conclusions change after applying this correction?

The larger or more complex a random experiment is, the more scope there is to *data snoop* to elucidate a "significant" finding. For example, after collecting the data above, we can notice that the biggest discrepancy in counts was between the "sixes" and the "threes". If we were to test $H_0 : p_6 = p_3$ we would find very strong evidence against H_0 in favour of $H_1 : p_6 \neq p_3$, and we may be led to (erroneously) conclude that the die is loaded.

We must always be honest about how we come up with our test hypotheses. If we choose them *after* looking at the data, then we must account for multiplicity.

Exercise 3 (Are suicide rates constant throughout the year?[1])

Are suicide rates constant throughout the year, or could they be affected by seasonal factors such as weather, holidays, and financial cycles? To investigate this issue, the National Center for Health Statistics collected the following data on the distribution of suicides in the United States by month in 1970:

Month	Number of Suicides	Days/Month
Jan.	1867	31
Feb.	1789	28
Mar.	1944	31
Apr.	2094	30
May.	2097	31
Jun.	1981	30
Jul.	1887	31
Aug.	2024	31
Sep.	1928	30
Oct.	2032	31
Nov.	1978	30
Dec.	1859	31
Total	23480	365

(a) What was the overall rate of suicides per day throughout the year 1970?

(b) If the suicide rate is indeed constant throughout the year, what counts of suicides would we have expected to see in each of the twelve months?
(c) Compute Pearson's χ^2 statistic for these data. What is the distribution of this test statistic under the null?
(d) Compute and interpret a p-value for testing whether the suicide rate is constant throughout the year. What is your conclusion?
(e) Look at the signs and magnitudes of the individual *relative deviations* $(\mathrm{O}_i - \mathrm{E}_i)/\sqrt{\mathrm{E}_i}$ for each month i. Is there any pattern in these deviations? Qualitatively summarise your findings.

[1] Adapted from Exercise 9.28 from *Mathematical Statistics & Data Analysis* (Rice, 1995).

Exercise 4 (Assessing authorship via word usage[2])

Probabilistic models of word counts can be used to measure literacy styles, and statistical techniques applied to such counts have been used to settle controversies about disputed authorship. An interesting account is given by Morton (1978), who also gave the following example:

When Jane Austen died, she left the novel *Sanditon* only partially completed, but she left an outline for the remainder. A highly literate admirer of Austen's works finished the novel, attempting to emulate Austen's style, and the hybrid was published. Morton (1978) counted the occurrences of various words in several works: chapters 1 and 3 of *Sense and Sensibility*, chapters 1, 2, and 3 of *Emma*, chapters 1 and 6 of *Sanditon* (written by Austen) and chapters 12 and 24 of *Sanditon* (written by her admirer). The counts Morton obtained for six words are given in the following table:

Word	*Sense and Sensibility*	*Emma*	*Sanditon (Austen)*	*Sanditon (Imitator)*
a	147	186	101	83
an	25	26	11	29
this	32	39	15	15
that	94	105	37	22
with	59	74	28	43
without	18	10	10	4
Total	375	440	202	196

(a) Are the relative frequencies of usage of these six words consistent across the three Austen works? Use a test of homogeneity to assess this.
(b) In light of your findings from part (a), carry out an appropriate test to assess whether the admirer was successful in imitating Austen's writing style in terms of usage rates of these six words.
(c) Look at the signs and magnitudes of the individual *relative deviations* $(\mathrm{O}_i - \mathrm{E}_i)/\sqrt{\mathrm{E}_i}$ for each of the six words. Which words were over or under used by the imitator? Qualitatively summarise your findings.

[2] Adapted from Section 13.3 of *Mathematical Statistics & Data Analysis* (Rice, 1995).